

On the uniqueness of flat thermodynamic geometries

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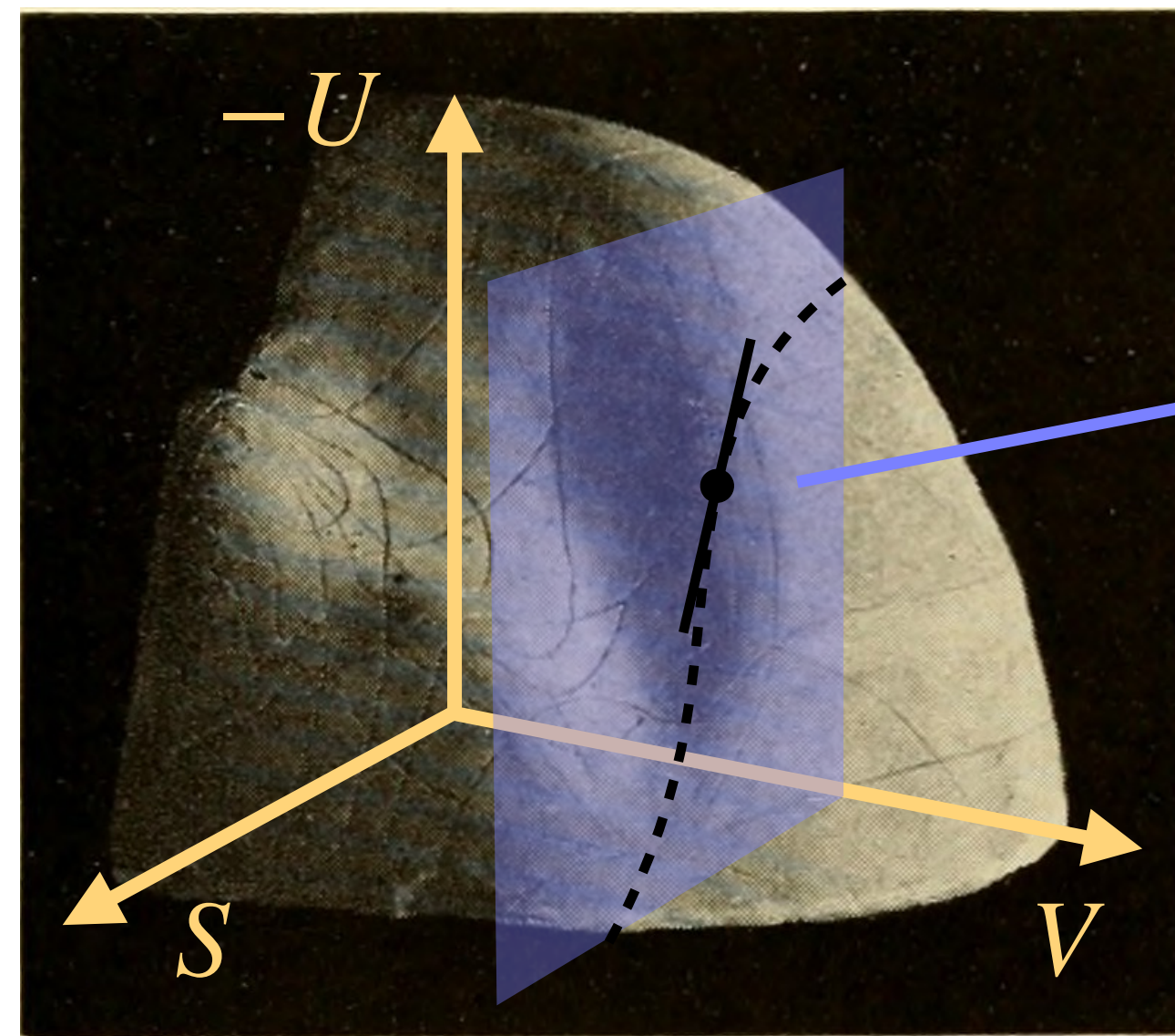
OUTLINE

- I. Early geometric ideas in thermodynamics
- II. Ruppeiner geometry
- III. Flat Ruppeiner metrics
- IV. Conclusions and recommendations

I. Early geometric ideas in thermodynamics

Early geometric ideas in thermodynamics

- Gibbs envisioned the equilibrium states of a system as existing on a "thermodynamic surface." Tangents to this surface encode additional information.

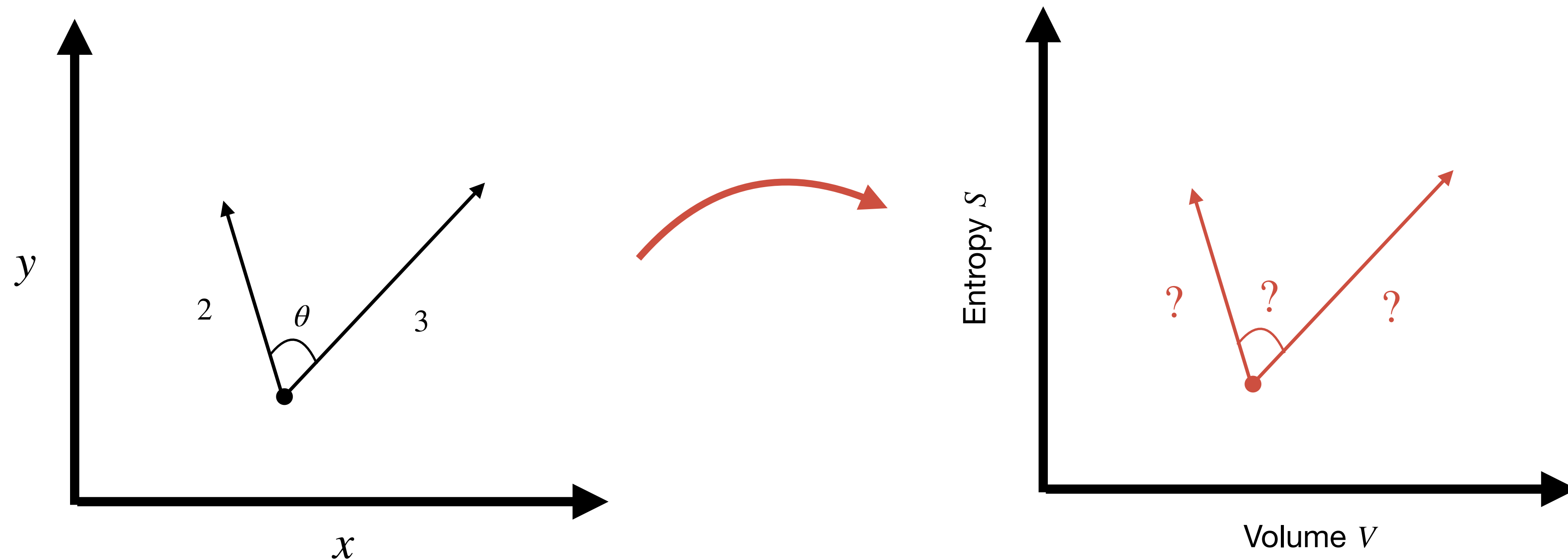


Slope at this point gives temperature, via $T = \left(\frac{\partial U}{\partial S} \right)_{V,N}$,
and similar interpretation can be attached to $-P = \left(\frac{\partial U}{\partial V} \right)_{S,N}$.

- An interesting shift in perspective, restating the fact that the fundamental relation $U = U(S, V, N)$ contains all the thermodynamic information of a system.

Early geometric ideas in thermodynamics

- Over a century later, Weinhold asks whether thermodynamic state space can be endowed with a physically meaningful metric, in order to obtain a notion of thermodynamic geometry.



- He finds that the answer is **yes**: one can draw a correspondence between the principles of thermodynamics and of metric geometry.

Early geometric ideas in thermodynamics

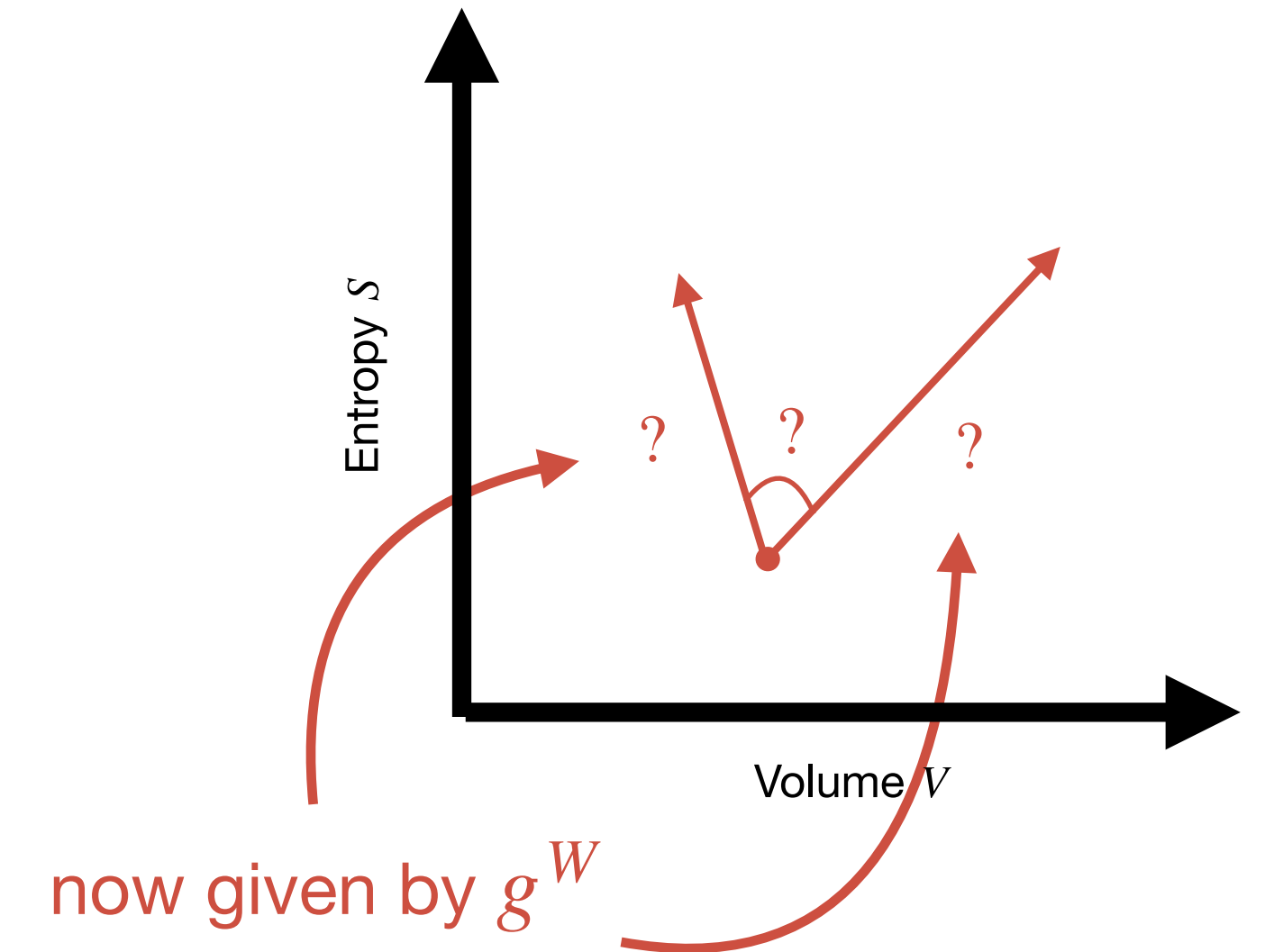
- In particular, Weinhold proposes a metric with components

$$g_{\alpha\beta}^W = \frac{\partial^2 U}{\partial X^\alpha \partial X^\beta},$$

where $X^\mu = (S, V, N)$. In Weinhold's formalism, vectors on the state space are associated with the equilibrium system's *response* to a change in a state variable X^α .

$$\mathcal{R}_\alpha := \partial U / \partial X^\alpha, \quad d\mathcal{R}_\alpha \leftrightarrow |\mathcal{R}_\alpha\rangle.$$

- The principles of thermodynamics then correspond to properties of a metric space; we will give a brief example.

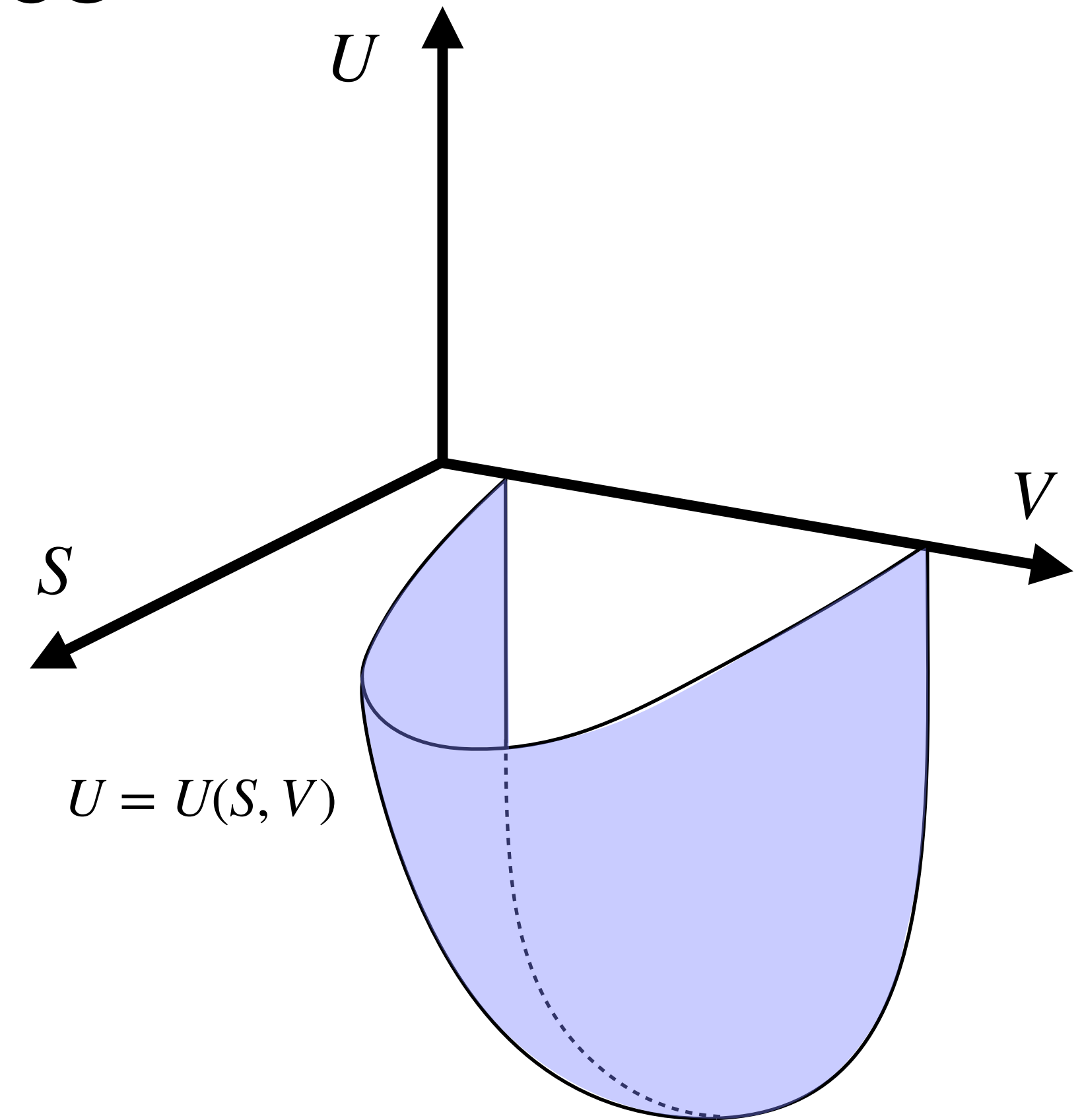


Early geometric ideas in thermodynamics

- The second law of thermodynamics says that U is minimized for equilibrium states. Mathematically, this means the energy surface $U = U(S, V, N)$ is convex, and

$$\left(\frac{\partial^2 U}{\partial S^2} \right)_{V,N} \geq 0, \quad \left(\frac{\partial^2 U}{\partial V^2} \right)_{S,N} \geq 0.$$

Weinhold showed that this implies $\langle \mathcal{R}_i | \mathcal{R}_i \rangle \geq 0$, where equality holds if and only if $|\mathcal{R}_i\rangle = 0$.

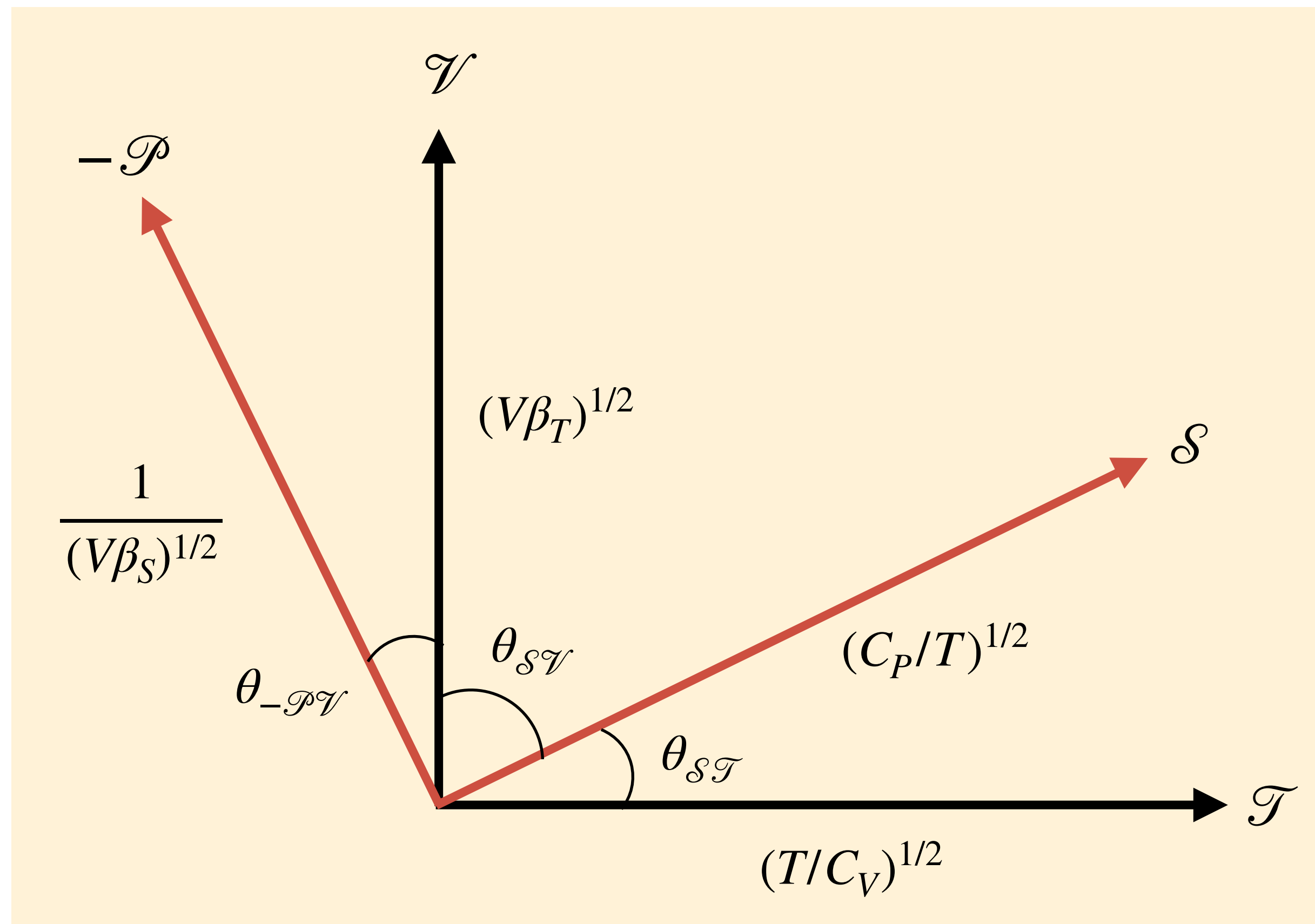


second law of thermodynamics $\leftrightarrow g^W$ is positive-definite

(Other properties of a metric space are also satisfied with g^W .)

Early geometric ideas in thermodynamics

- Weinhold's formalism, aside from providing a notion of lengths and angles, allows us to use the rules of plane geometry to obtain nontrivial thermodynamic identities.



The equality of cosines

$$\cos^2 \theta_{\mathcal{S}T} = \cos^2 \theta_{-\mathcal{P}V}$$

implies the thermodynamic identity

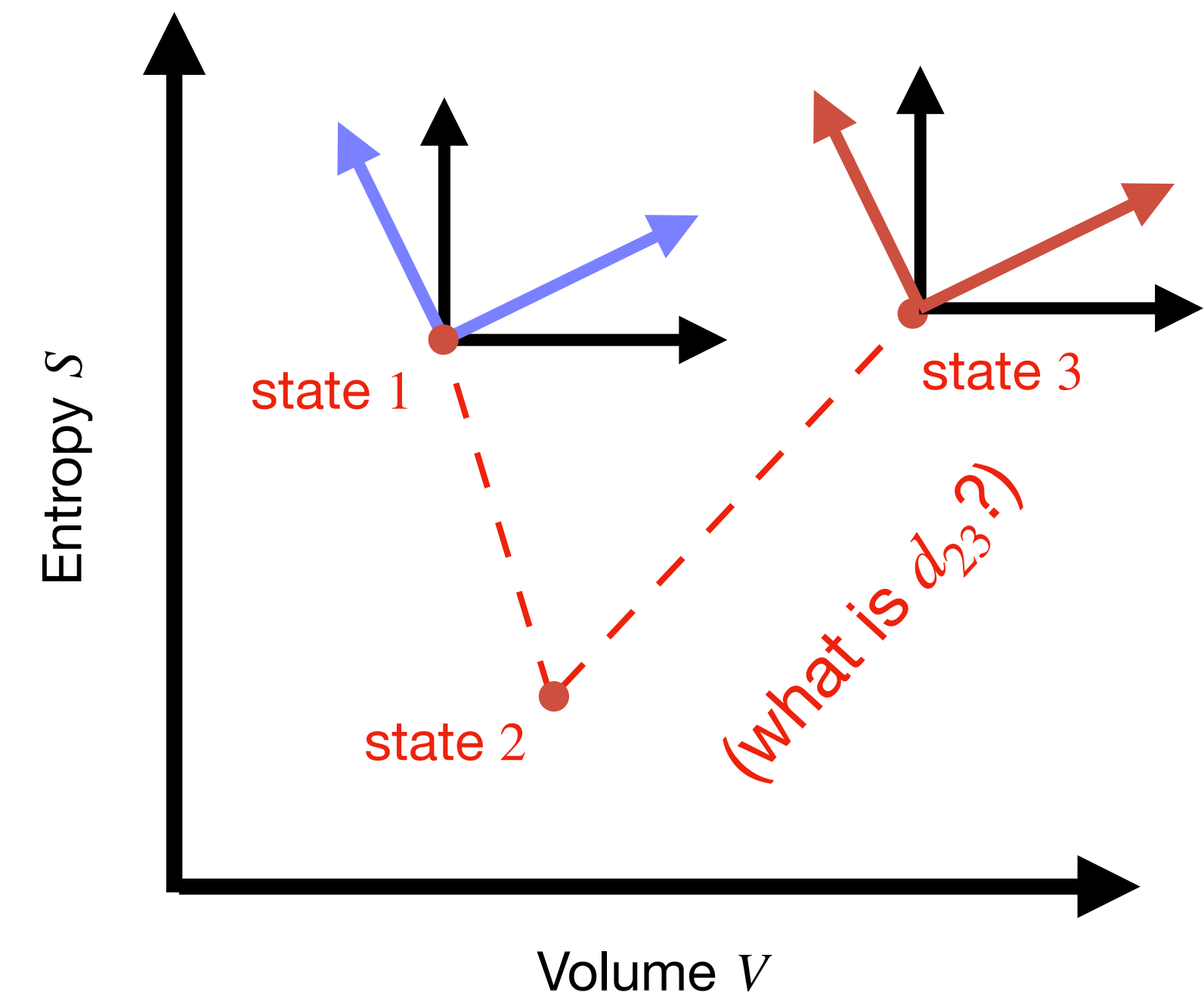
$$C_P/C_V = \beta_T/\beta_S.$$

Meanwhile, the geometric requirement $\cos^2 \theta_{\mathcal{S}T} \leq 1$ implies that $C_P \geq C_V$.

Early geometric ideas in thermodynamics

- Points for improvement / further exploration in Weinhold's thermodynamic geometry?

- (1) ***Designed to treat isolated equilibrium states.*** How to compare vectors corresponding to different states?
- (2) Weinhold's theory lacks a ***physically-motivated notion of distance between equilibrium states.***
- (3) A metric yields more than a notion of lengths and angles – it also defines ***geodesics and curvature.***



- Can these concerns be addressed by a different theory, perhaps a different metric?

II. Ruppeiner geometry

Ruppeiner geometry

- The Ruppeiner metric is motivated by thermodynamic fluctuation theory, which asks the question,

Given a system whose equilibrium state is A_0 , what is the probability for the system to be at some other state A in the vicinity of A_0 ?

The answer is given by

$$P(A_0 \rightarrow A) = C \exp \left[-\frac{1}{2k} \left(-\frac{\partial^2 S}{\partial X^\alpha \partial X^\beta} \right) \Big|_{A_0} \Delta X_A^\alpha \Delta X_A^\beta \right],$$

where $X^\mu = (U, V, N)$ are the natural variables of S and $\Delta X_A^\alpha \equiv X_A^\alpha - X_{A_0}^\alpha$.

Ruppeiner geometry

- Ruppeiner found that this result defines a metric with a physically-motivated notion of distance. In particular, the choice

$$g_{\alpha\beta}^R = - \frac{\partial^2 S}{\partial X^\alpha \partial X^\beta},$$

"distance" given by the metric

where $X^\mu = (U, V, N)$, allows us to make the identification

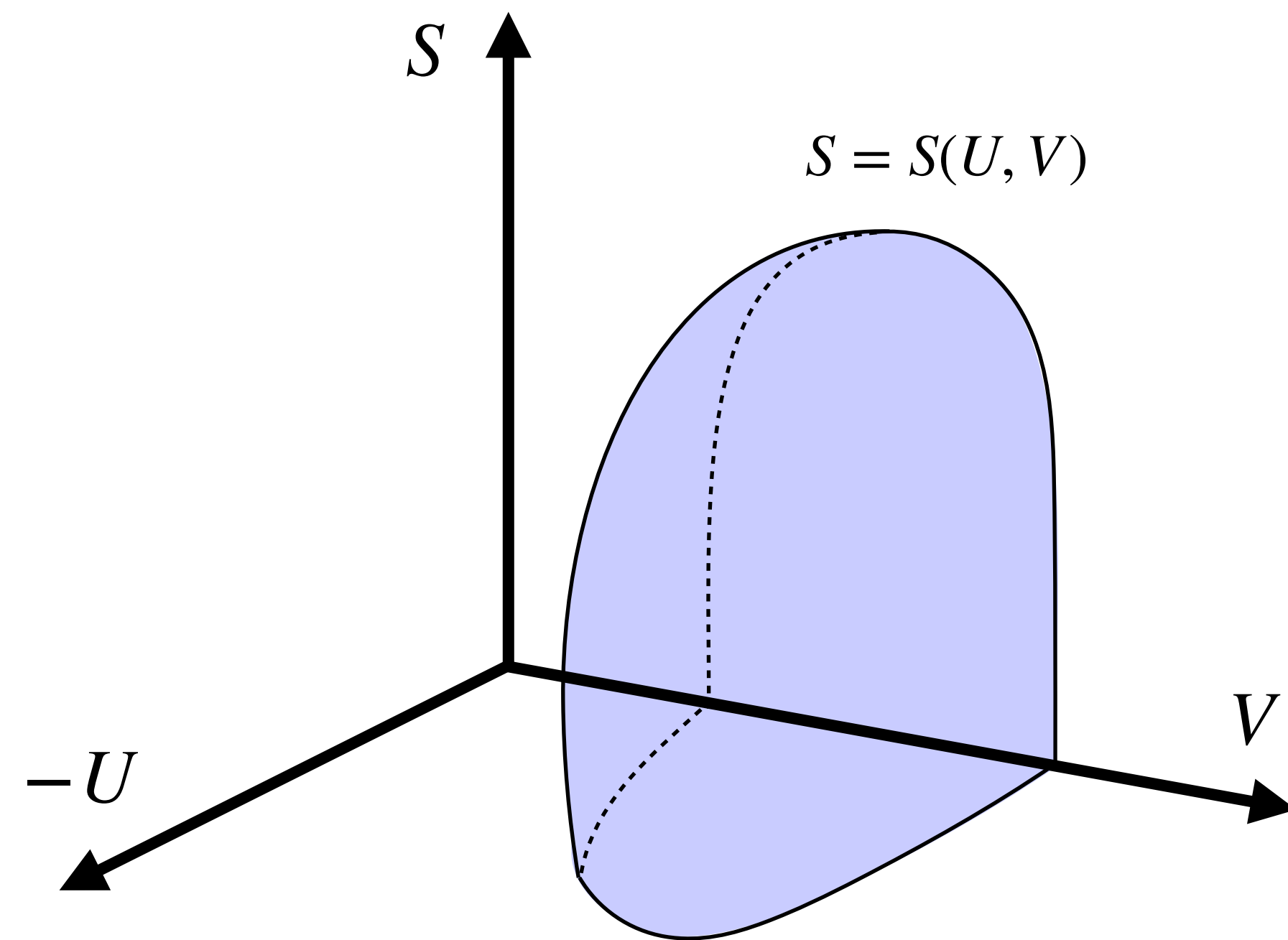
$$P(A_0 \rightarrow A) = C \exp \left[-\frac{1}{2k} g(\Delta \mathbf{X}_A, \Delta \mathbf{X}_A) \Big|_{A_0} \right].$$

- Thus, the "further apart" two states are, the less likely for a fluctuation to occur between them.

Ruppeiner geometry

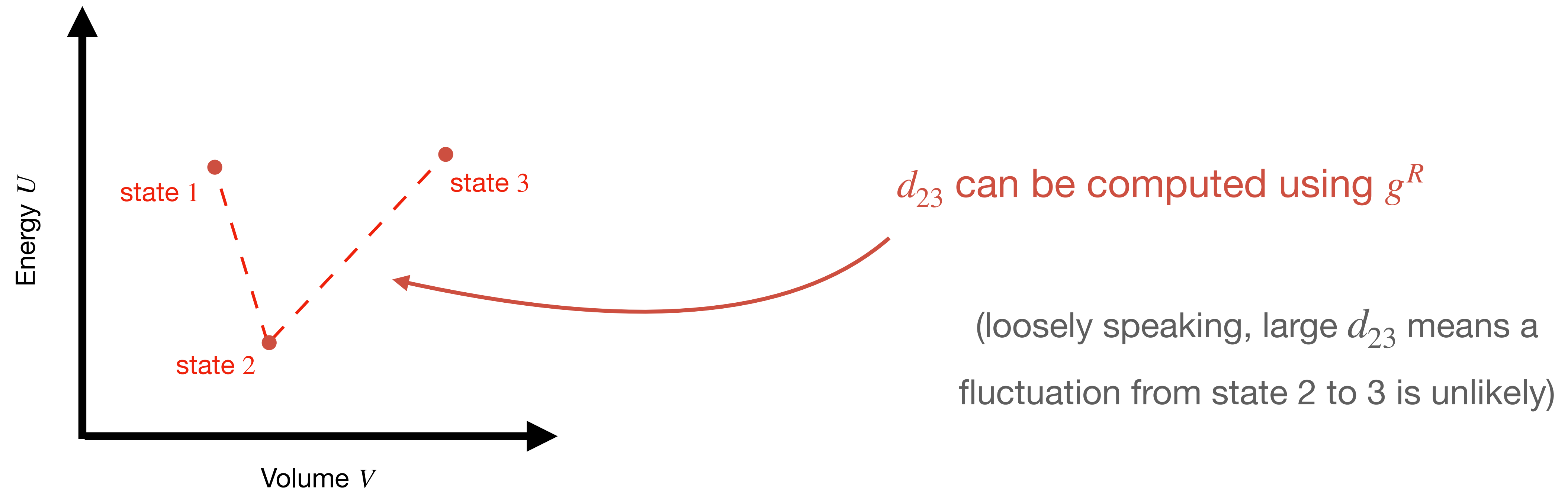
- Similar to the Weinhold metric, the second law of thermodynamics implies that the Ruppeiner metric g^R is positive-definite.
- The main difference is that entropy is *maximized* for equilibrium states, so that the entropy surface $S = S(U, V, N)$ is concave. This then implies

$$\left(\frac{\partial^2 S}{\partial U^2} \right)_{V,N} \leq 0, \quad \left(\frac{\partial^2 S}{\partial V^2} \right)_{U,N} \leq 0.$$



Ruppeiner geometry

- We now have some notion of what it means for thermodynamic states to be "near" or "far," distinguishing Ruppeiner's theory from Weinhold's.

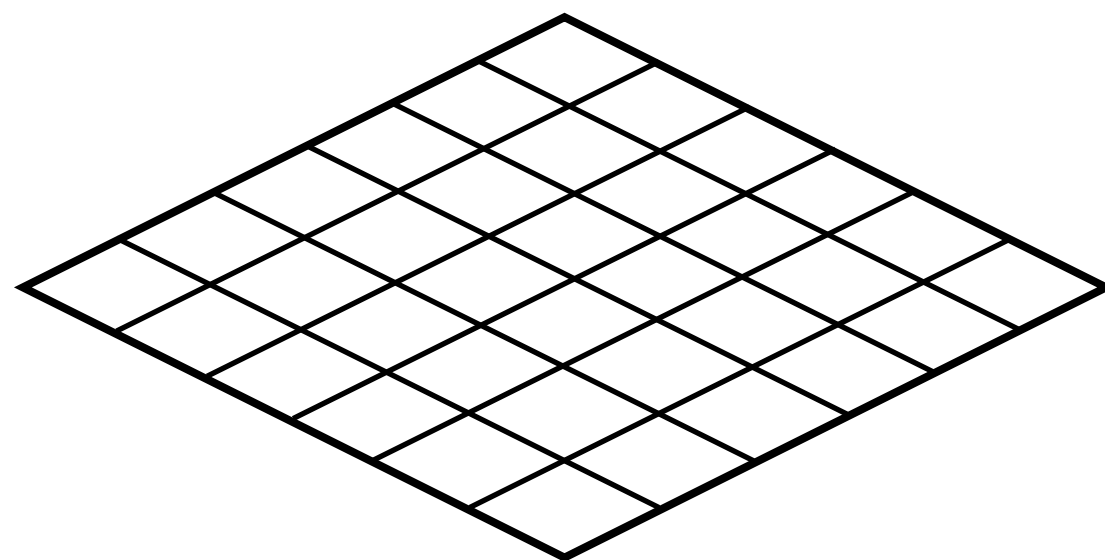


Ruppeiner geometry

- Ruppeiner's next insight was to look at curvature, paying special attention to the scalar curvature R :

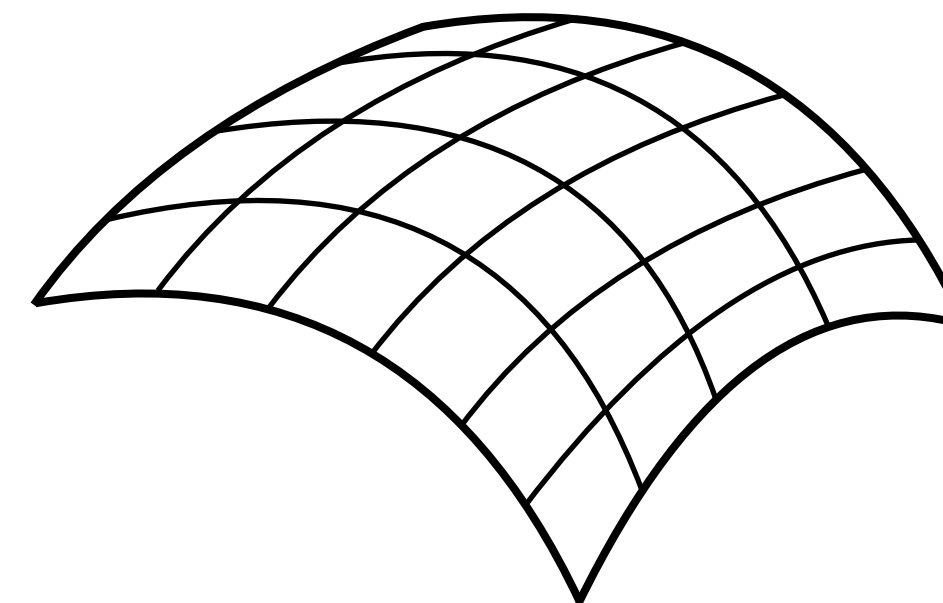
$$g_{\alpha\beta} \rightarrow R^\mu_{\nu\rho\sigma} \rightarrow R$$

- A direct calculation reveals that $R = 0$ for the ideal gas. Since this gas is characterized by being composed of non-interacting particles, this led Ruppeiner to conjecture that ***curvature indicates the presence of interactions.***



$$R = 0$$

no interactions



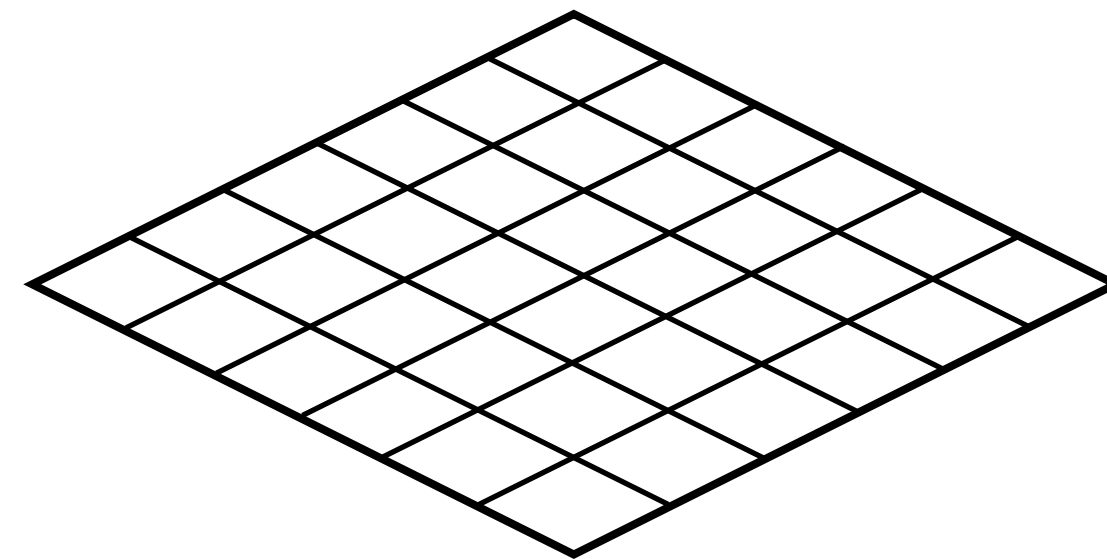
$$R \neq 0$$

nontrivial interactions

Ruppeiner geometry

- In going from Weinhold to Ruppeiner geometry, a primary concern was the absence of a physically-motivated distance between equilibrium states.
- Unexpectedly, the metric g^R appears to provide much more than a notion of distance, due to Ruppeiner's conjecture.

What are the limitations of this conjecture?



$$R = 0$$

no interactions

III. Flat Ruppeiner metrics

Flat Ruppeiner metrics

- First, it's not very convenient to work in terms of the natural variables $X^\mu = (U, V, N)$ of S ; we rewrite the Ruppeiner metric in terms of the variables (T, V, N) .

Second, a necessary step in the derivation of the fluctuation theory result for $P(A_0 \rightarrow A)$ is that one extensive variable must be held fixed; this sets a scale for the system.

- When volume V is held constant, we get

$$g_V = -\frac{1}{T} \left(\frac{\partial^2 F}{\partial T^2} \right) dT^2 + \frac{1}{T} \left(\frac{\partial^2 F}{\partial N^2} \right) dN^2,$$

while for constant N we have

$$g_N = -\frac{1}{T} \left(\frac{\partial^2 F}{\partial T^2} \right) dT^2 + \frac{1}{T} \left(\frac{\partial^2 F}{\partial V^2} \right) dV^2.$$

Flat Ruppeiner metrics

- There are *two* Ruppeiner metrics, corresponding to whether V or N is fixed. We call these the *Ruppeiner- V and N metrics*, respectively.
- The former is used almost exclusively in the literature (it is the one studied by Ruppeiner), but the Ruppeiner- N metric seems to be better suited for the analysis of phase transitions.
- Why care? ***Ruppeiner's conjecture is sensitive to the choice of metric.*** Is it $R_V = 0$ or $R_N = 0$ which indicates the absence of interactions?

$$g_V \rightarrow R_V, \quad g_N \rightarrow R_N.$$

Flat Ruppeiner metrics

- A metric $g_{\alpha\beta}$ that yields a vanishing Riemann curvature tensor $R^\mu_{\nu\rho\sigma}$ is called flat.

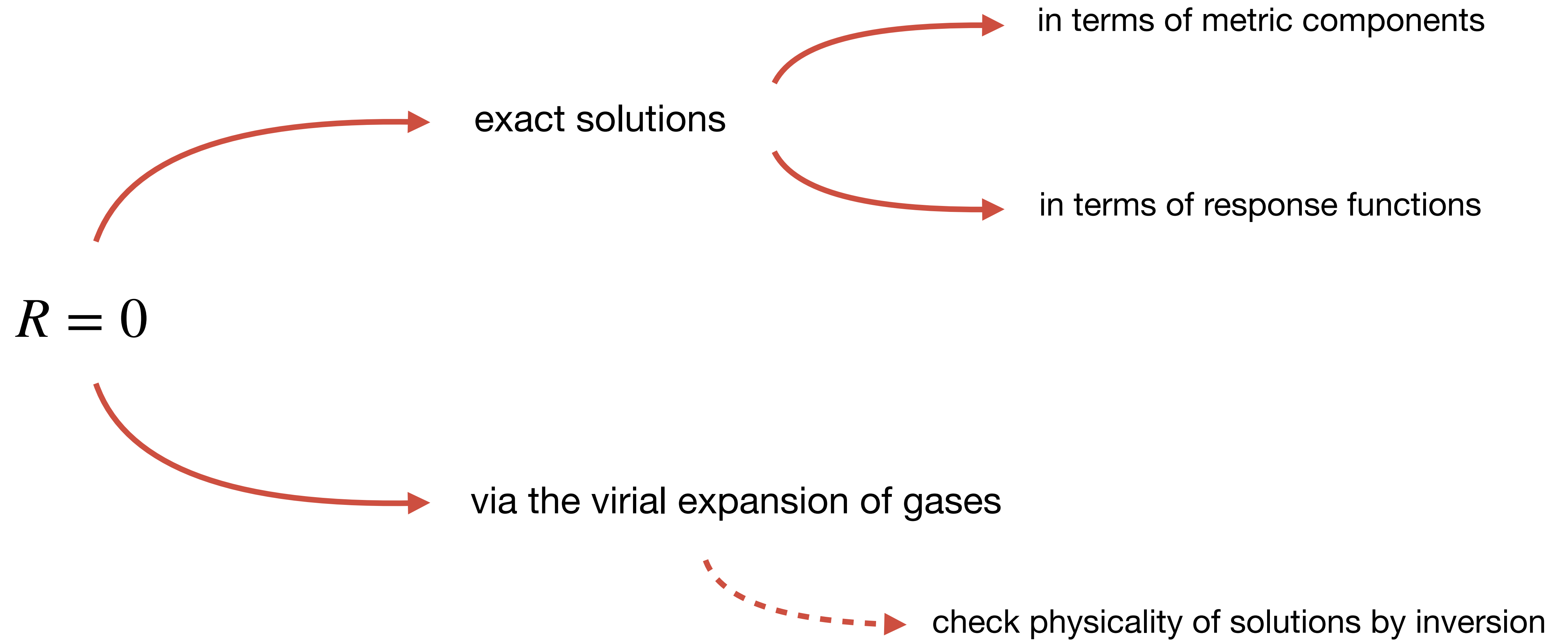
In $d = 2$ dimensions, $R^\mu_{\nu\rho\sigma} \propto R$, so that $R = 0$ implies a flat metric.

- From the perspective of Ruppeiner's theory, flat metrics must correspond to non-interacting systems. ***Can we construct a flat metric which corresponds to an interacting system?***
- This is difficult, because finding solutions to $R = 0$ involves solving a high-order PDE.

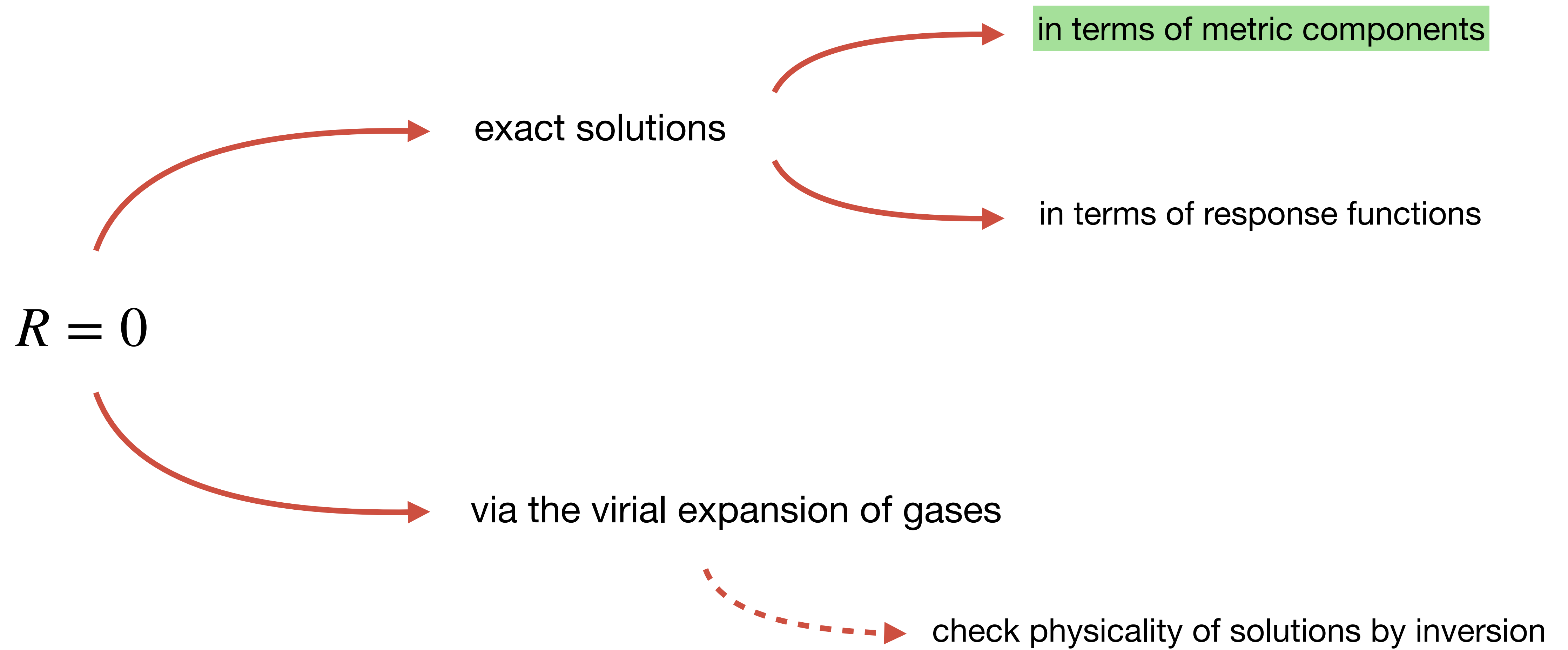
To give a sketch for the Ruppeiner- V metric, we have

$$R_V = \frac{1}{\sqrt{g_{TT}g_{NN}}} \left(\frac{\partial}{\partial T} \left(\frac{1}{\sqrt{g_{TT}g_{NN}}} \frac{\partial g_{NN}}{\partial T} \right) + \frac{\partial}{\partial N} \left(\frac{1}{\sqrt{g_{TT}g_{NN}}} \frac{\partial g_{TT}}{\partial N} \right) \right).$$

Constructing flat Ruppeiner metrics



Constructing flat Ruppeiner metrics



Flat Ruppeiner metrics

Exact solutions (metric components)

- In the coordinates (T, V, N) , the Ruppeiner metric is diagonal and takes the form

$$g = g_{TT} dT^2 + g_{XX} dX^2,$$

where the choice of $X = N$ corresponds to the Ruppeiner- V metric, and $X = V$ gives Ruppeiner- N .

The curvature scalar is then given by

$$R_X = \frac{1}{\sqrt{g_{TT}g_{XX}}} \left(\frac{\partial}{\partial T} \left(\frac{1}{\sqrt{g_{TT}g_{XX}}} \frac{\partial g_{XX}}{\partial T} \right) + \frac{\partial}{\partial X} \left(\frac{1}{\sqrt{g_{TT}g_{XX}}} \frac{\partial g_{TT}}{\partial X} \right) \right).$$

Flat Ruppeiner metrics

Exact solutions (metric components)

- To simplify the expression for the curvature scalar, we require that $\partial g_{TT}/\partial X = 0$. Then

$$R_X = \frac{1}{\sqrt{g_{TT}g_{XX}}} \frac{\partial}{\partial T} \left(\frac{1}{\sqrt{g_{TT}g_{XX}}} \frac{\partial g_{XX}}{\partial T} \right),$$

and then it can be shown that $R_X = 0$ if and only if

$$g_{XX} = \left(A(X) \int dT \sqrt{g_{TT}} + B(X) \right),$$

where A and B are free functions of X .

Flat Ruppeiner metrics

Exact solutions (metric components)

- To restate this in more physical terms, we recall that the metric components are defined by partial derivatives of the Helmholtz free energy F , which can be related to *response functions*:

$$g_V = -\frac{1}{T} \left(\frac{\partial^2 F}{\partial T^2} \right) dT^2 + \frac{1}{T} \left(\frac{\partial^2 F}{\partial N^2} \right) dN^2,$$

$$C_V \equiv \left(\frac{\partial U}{\partial T} \right)_V = -T \left(\frac{\partial^2 F}{\partial T^2} \right)$$

(isochoric heat capacity)

$$g_N = -\frac{1}{T} \left(\frac{\partial^2 F}{\partial T^2} \right) dT^2 + \frac{1}{T} \left(\frac{\partial^2 F}{\partial V^2} \right) dV^2.$$

$$B_T \equiv -V \left(\frac{\partial P}{\partial V} \right)_T = V \left(\frac{\partial^2 F}{\partial V^2} \right)$$

(isothermal bulk modulus)

- That is, the Ruppeiner metric can be written entirely in terms of response functions.

Flat Ruppeiner metrics

Exact solutions (metric components)

- Eliminating metric components in favor of response functions, we obtain the ff. pair of results:
- The Ruppeiner- N metric is flat for any system with $\partial C_V / \partial V = 0$ and

$$B_T = T \left(A(V) + B(V) \int dT \frac{\sqrt{C_V}}{T} \right)^2,$$

while the Ruppeiner- V metric is flat for any system with $\partial C_V / \partial N = 0$ and

$$\left(\frac{\partial \mu}{\partial N} \right)_{T,V} = T \left(A(N) + B(N) \int dT \frac{\sqrt{C_V}}{T} \right)^2.$$

- Immediate conclusion: there are non-ideal gas systems with vanishing scalar curvature!

Flat Ruppeiner metrics

Exact solutions (metric components)

- Eliminating metric components in favor of response functions, we obtain the ff. pair of results:

- The Ruppeiner- N metric is flat for any system with $\partial C_V / \partial V = 0$ and

$$B_T = T \left(A(V) + B(V) \int dT \frac{\sqrt{C_V}}{T} \right)^2,$$

ideal gas is the special case
 $A(V) = Nk/V$ and $B(V) = 0$

while the Ruppeiner- V metric is flat for any system with $\partial C_V / \partial N = 0$ and

$$\left(\frac{\partial \mu}{\partial N} \right)_{T,V} = T \left(A(N) + B(N) \int dT \frac{\sqrt{C_V}}{T} \right)^2.$$

problematic!

- Immediate conclusion: there are non-ideal gas systems with vanishing scalar curvature!

Flat Ruppeiner metrics

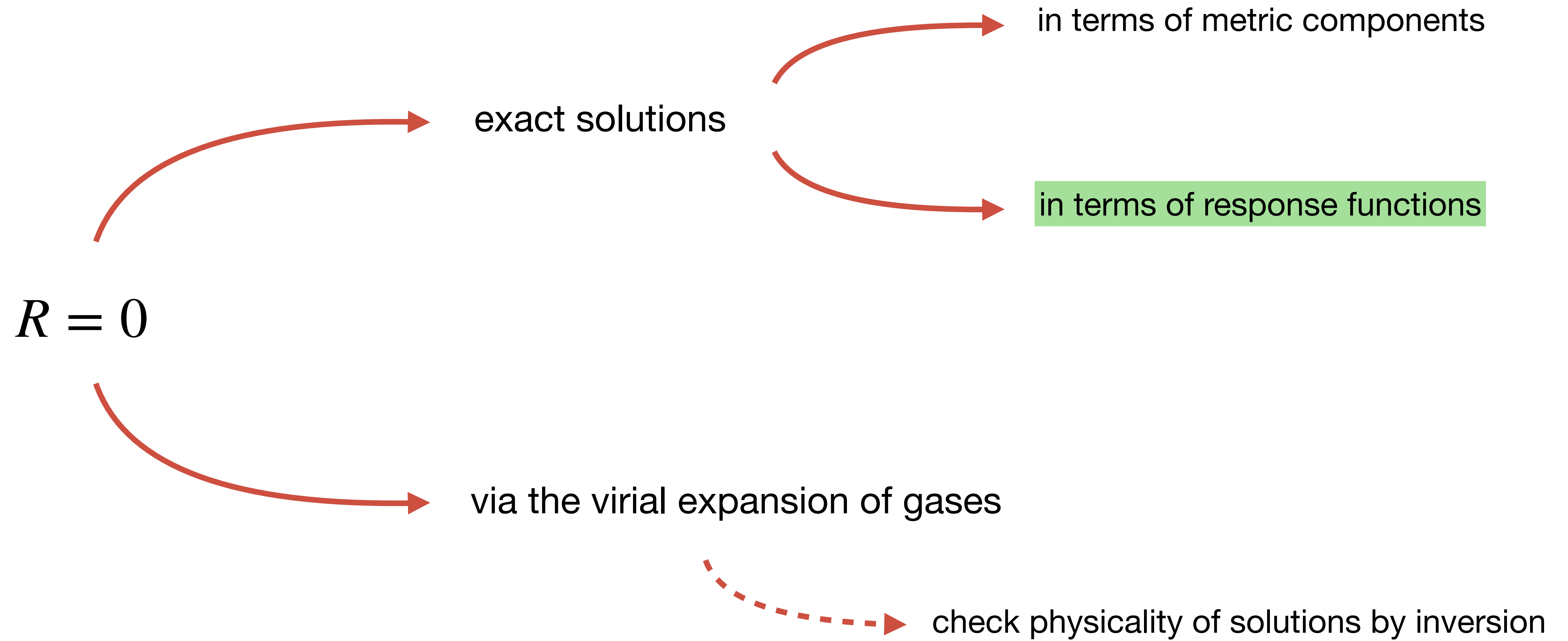
A problem with our assumptions

- The constraint $\partial C_V / \partial N = 0$, which we imposed in order to find flat Ruppeiner- V metrics, seems generally inapplicable to physical systems:

One expects that the energy needed to change the temperature of a gas has a nontrivial dependence on the number of particles N in the gas, and since $C_V := (\partial U / \partial T)_V$, this amounts to the expectation that $\partial C_V / \partial N \neq 0$.

- To address this issue, we try a different approach.

Constructing flat Ruppeiner metrics



Flat Ruppeiner metrics

Exact solutions (response functions)

- The Ruppeiner metric is given in terms of derivatives of F , which can, in turn, be expressed in terms of thermodynamic response functions. For the Ruppeiner- V metric, we have

$$R_V^* = \frac{1}{B^*C^*} \left\{ T \frac{\partial B^*}{\partial T} \left[1 - \frac{1}{2} T \frac{\partial \ln(B^*C^*)}{\partial T} \right] + T^2 \frac{\partial^2 B^*}{\partial T^2} \right. \\ \left. + \rho \frac{\partial C^*}{\partial \rho} \left[1 - \frac{1}{2} \rho \frac{\partial \ln(B^*C^*)}{\partial \rho} \right] + \rho^2 \frac{\partial^2 C^*}{\partial \rho^2} - \frac{1}{2} \rho C^* \frac{\partial \ln(B^*/C^*)}{\partial \rho} \right\}. \quad (\rho \equiv N/V)$$

Starred quantities have been nondimensionalized, and the response functions normalized to those of the ideal gas, so that $R_V^* := R_V \rho$, $C^* := C_V / Nk$, and $B^* := B_T / \rho kT$.

- Since C^* and B^* are constant for the ideal gas, their derivatives vanish, and $R_V^* = 0$, as expected.

Flat Ruppeiner metrics

Exact solutions (response functions)

- To simplify our analysis, we consider the class of thermodynamic systems whose isochoric heat capacity is of the form $C_V = Nk f(T)$, where $f(T)$ is some positive definite function. Notably, $\partial C_V / \partial N \neq 0$; we have evaded the unphysical assumption used earlier. This leaves only the isothermal bulk modulus B_T to be determined:

$$R_V^* = \frac{1}{B^* C^*} \left\{ T \frac{\partial B^*}{\partial T} \left[1 - \frac{1}{2} T \frac{\partial \ln(B^* C^*)}{\partial T} \right] + T^2 \frac{\partial^2 B^*}{\partial T^2} \right. \\ \left. + \rho \frac{\partial C^*}{\partial \rho} \left[1 - \frac{1}{2} \rho \frac{\partial \ln(B^* C^*)}{\partial \rho} \right] + \rho^2 \frac{\partial^2 C^*}{\partial \rho^2} - \frac{1}{2} \rho C^* \frac{\partial \ln(B^* / C^*)}{\partial \rho} \right\}.$$

- This equation is still very difficult to solve for general B^* , so we make further assumptions about the functional form of the (normalized) bulk modulus.

Flat Ruppeiner metrics

ASSUMPTIONS

RESULTING SOLUTIONS TO $R_V^ = 0$*

(1) $B^* = B^*(T)$ and $C_V \sim Nk$

$$B^* = c_2(c_1 + \ln T)^2.$$

(2) $B^* = B^*(\rho)$

$$B^* = \text{constant}$$

(3) $B^*(T, \rho) = \Theta(T) R(\rho)$

Generally complex-valued, unless separation constant A is zero. Then if $A = 0$ and $C_V \sim Nk$, we recover solution (1).

Flat Ruppeiner metrics

ASSUMPTIONS

RESULTING SOLUTIONS TO $R_V^* = 0$

(1) $B^* = B^*(T)$ and $C_V \sim Nk$

$$B^* = c_2(c_1 + \ln T)^2.$$

 non-ideal, since non-constant

(2) $B^* = B^*(\rho)$

$$B^* = \text{constant}$$

(3) $B^*(T, \rho) = \Theta(T) R(\rho)$

Generally complex-valued, unless separation constant A is zero. Then if $A = 0$ and $C_V \sim Nk$, we recover solution (1).

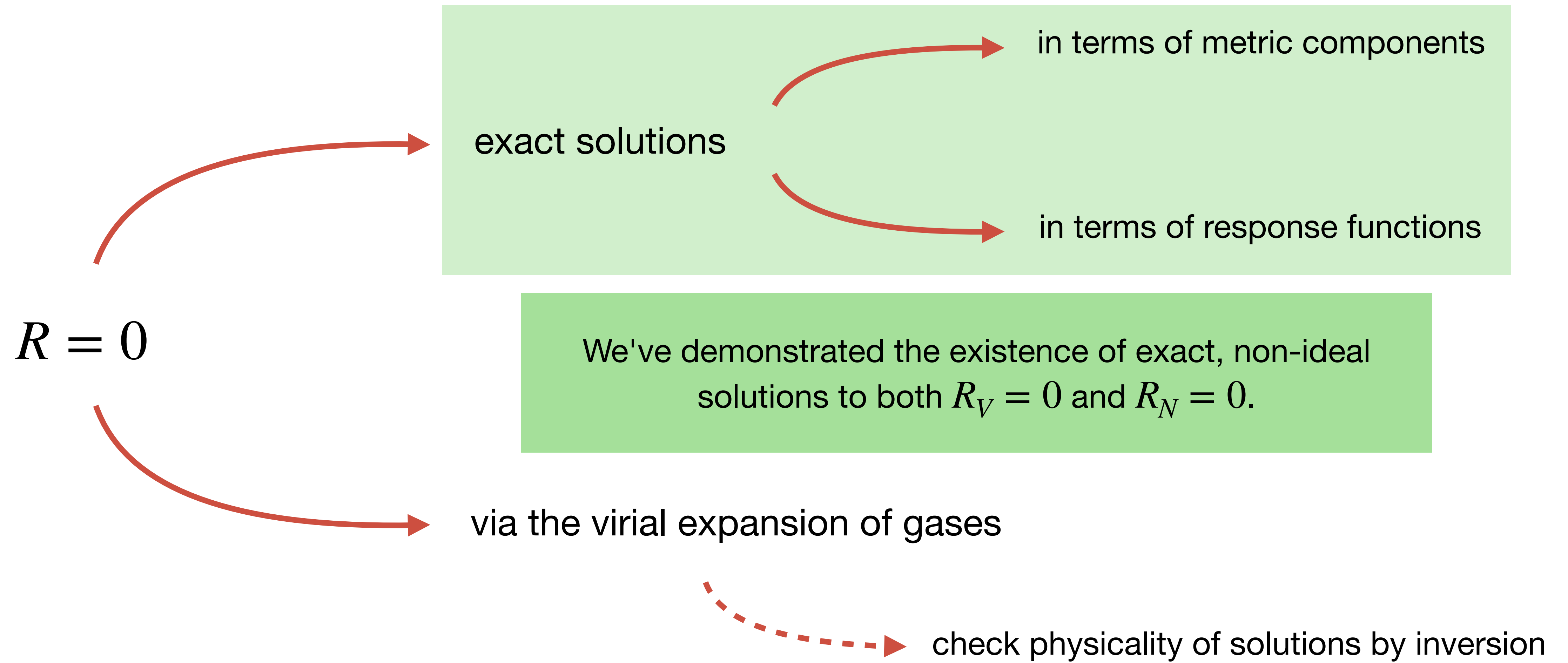
Flat Ruppeiner metrics

Exact solutions (response functions)

- To recap, the response-function approach allows us to conclude that exact, non-ideal gas solutions to $R_V = 0$ exist, as exemplified by solution (1) on the previous slide.

Moreover, we can make this claim without needing to assume that $\partial C_V / \partial N = 0$, which we used in the metric-component approach.

Constructing flat Ruppeiner metrics



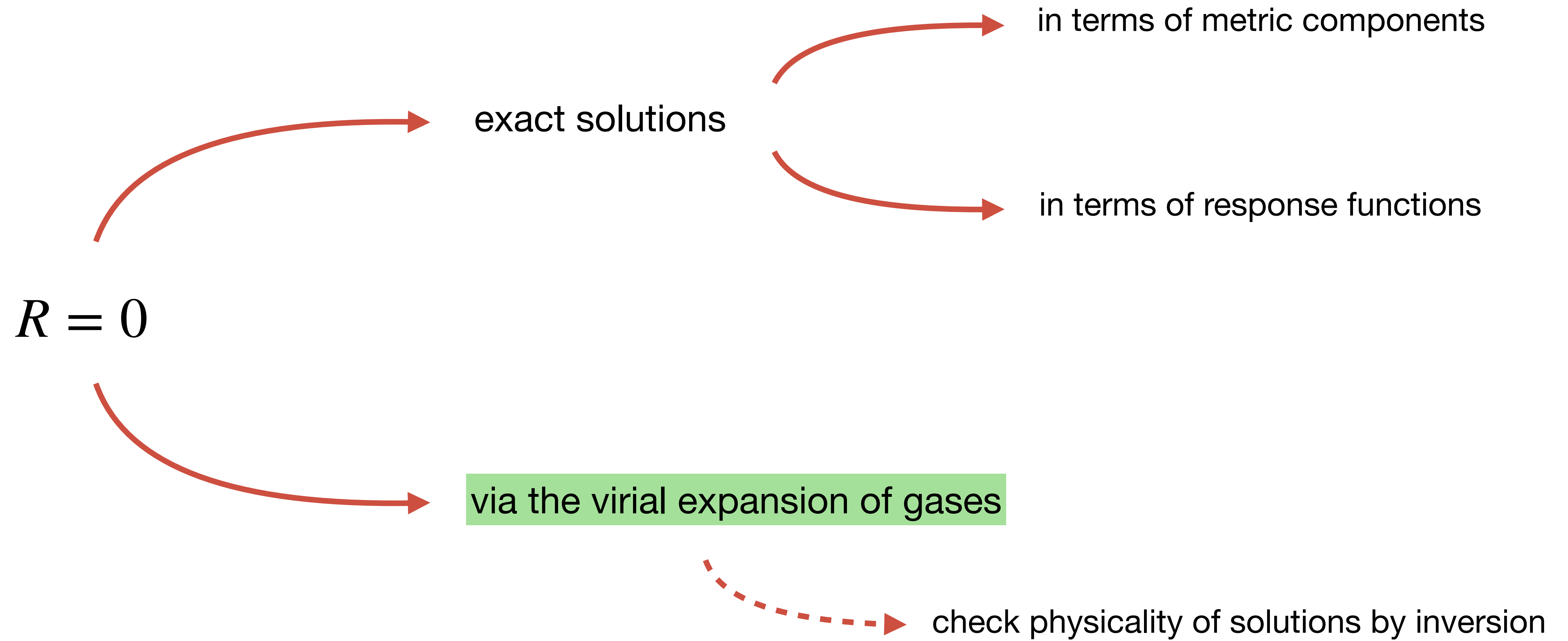
Flat Ruppeiner metrics

- If we want our exact solutions to make direct contact with Ruppeiner's conjecture, we must make an assumption to the effect of

If the thermodynamic response functions of the system differ from those of the ideal gas, then the system exhibits nontrivial interactions.

- More convincing proof that there are flat solutions with nontrivial interactions?

Constructing flat Ruppeiner metrics



Flat Ruppeiner metrics

Using the virial expansion

- A new starting-point for our analysis is provided by the *virial expansion*

$$F(T, V, N) = NkT \left(\ln \left(\frac{N}{V} \right) - \frac{3}{2} \ln(\gamma T) - 1 + B_2(T) \frac{N}{V} + \frac{1}{2} B_3(T) \left(\frac{N}{V} \right)^2 + \dots \right),$$

which is used to describe gases in the low density, high temperature regime. It fixes the functional dependence of F on N and V , leaving only the *virial coefficients* $B_2, B_3, \dots$ to be specified.

- Crucially, the virial coefficients $B_2, B_3, \dots$ can be directly linked to the nature of the intermolecular potential of the system.
- Note that $B_i = 0$ for every i in the case of the ideal gas.

Flat Ruppeiner metrics

Using the virial expansion

- Schematically, the calculation goes $F \rightarrow g_{\alpha\beta} \rightarrow R$, with the result

$$\begin{aligned} R_V = & \frac{1}{3V} \left(T^2 \frac{d^2 B_2}{dT^2} - 2T \frac{dB_2}{dT} - 3B_2 \right) \\ & + \frac{1}{9V} \left(-20T^2 \left(\frac{dB_2}{dT} \right)^2 + 24T \frac{dB_2}{dT} B_2 + 4T^4 \frac{d^3 B_2}{dT^3} \frac{dB_2}{dT} + 8T^3 \frac{d^2 B_2}{dT^2} \frac{dB_2}{dT} \right. \\ & \left. + 36B_2^2 + 3T^2 \frac{d^2 B_3}{dT^2} - 12T \frac{dB_3}{dT} - 27B_3 \right) \rho \\ & + O(\rho^2), \end{aligned}$$

and the expansion for R_N is computed similarly. Rather than demand an exact solution, we can require the vanishing of the coefficients of ρ in this expansion.

Flat Ruppeiner metrics

Using the virial expansion

- If we demand that R_V vanishes to lowest order in ρ , then we must have

$$B_2(T) = c_1 T^{(3-\sqrt{21})/2} + c_2 T^{(3+\sqrt{21})/2},$$

where c_1 and c_2 are constants of integration, while for R_N we obtain

$$B_2(T) = c_3 + c_4 T.$$

- These virial coefficients are generally distinct (i.e. cannot correspond to the same system) unless $c_1 = c_2 = c_3 = c_4 = 0$, which corresponds to the ideal gas. This suggests that

The ideal gas is the only system for which both R_V and R_N vanish to lowest order in ρ .

Flat Ruppeiner metrics

Using the virial expansion

- We also attempt to find virial coefficients such that R_V (or R_N) vanishes to *all* orders in ρ .
- In order for the Ruppeiner- V metric to vanish to high order in ρ , we choose the coefficients

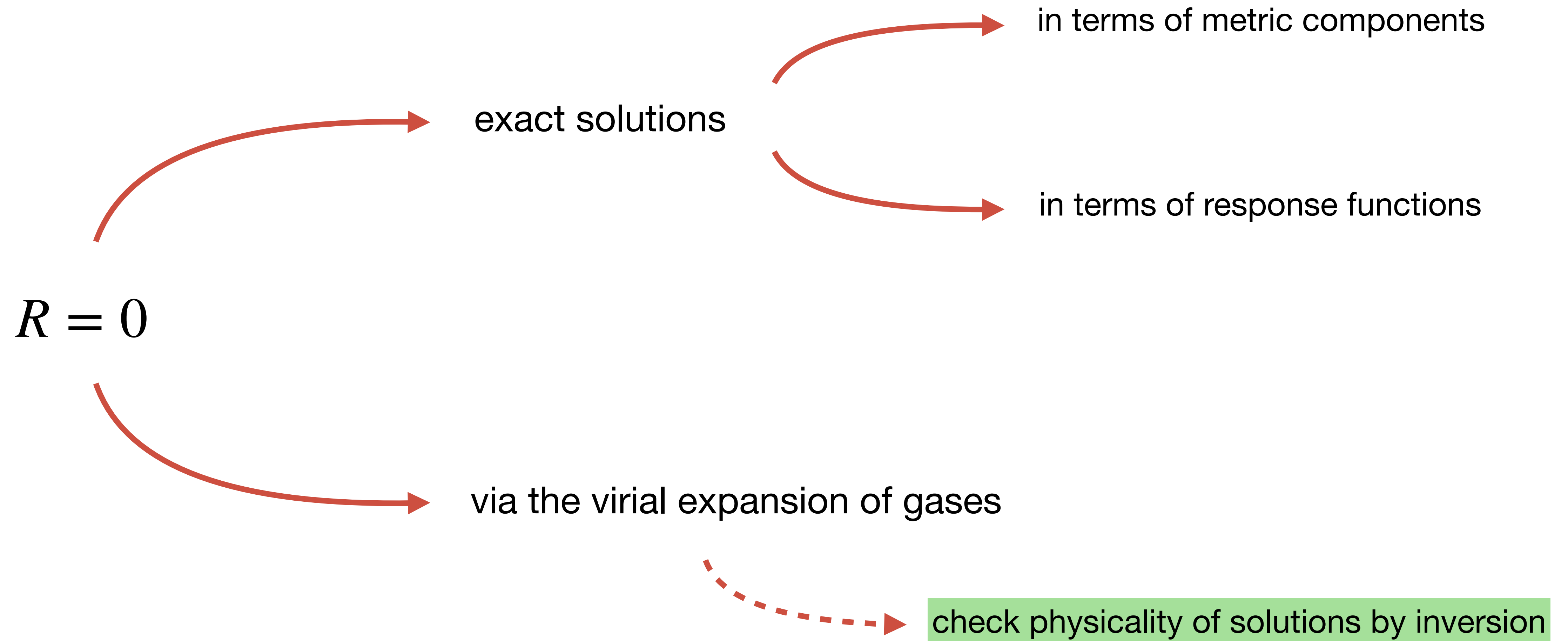
$$B_n(T) = \bar{B}_n \alpha^{n-1} T^{(3-\sqrt{21})(n-1)/2},$$

where $\bar{B}_n$ are constants whose values must be determined by direct calculation. Meanwhile, for the Ruppeiner- N metric, the choice

$$B_n(T) = \bar{B}_n,$$

where $\bar{B}_n$ can be *any* constants, yields $R_N \rightarrow 0$.

Constructing flat Ruppeiner metrics



Flat Ruppeiner metrics

Inversion of virial coefficients

- Assuming the interactions between particles are sufficiently simple (i.e. pairwise, with no angular dependence), the second virial coefficient is given by

$$B_2(\beta) = 2\pi \int_0^\infty r^2 \left(1 - e^{\beta u(r)}\right) dr,$$

where $\beta \equiv 1/kT$ and $u(r)$ is the intermolecular potential as a function of separation r between molecules.

- We develop an inversion procedure which works for monotonic potentials $u(r)$, so that given a second virial coefficient B_2 , we can determine the corresponding inter-particle interactions.

Flat Ruppeiner metrics

Inversion of virial coefficients

- By specifying the limiting behavior of $u(r)$ and assuming a monotonic potential, we show that

$$B_2(\beta) = 2\pi \int_0^\infty r^2 (1 - e^{\beta u(r)}) dr \quad \Longrightarrow \quad r^3 = \frac{3}{2\pi} L^{-1} \left(\frac{B_2(\beta)}{\beta} \right),$$

where L^{-1} denotes an inverse Laplace transform. Since the right-hand side of our inversion formula is a function of u alone, it can (in principle) be inverted to obtain u as a function of r .

Flat Ruppeiner metrics

Inversion of virial coefficients

- We recall our solutions from earlier:

$$B_2(T) = c_1 T^{(3-\sqrt{21})/2} + c_2 T^{(3+\sqrt{21})/2} \quad (R_V \rightarrow 0)$$

$$B_2(T) = c_3 + c_4 T \quad (R_N \rightarrow 0)$$

- With the use of our inversion procedure, we show that the $R_V \rightarrow 0$ solution can be made to correspond to a particular inverse-power potential, $u(r) = \frac{\gamma}{r^{(3+\sqrt{21})/2}}$.

Meanwhile, the $R_N \rightarrow 0$ solution can be made to correspond to a hard-sphere interaction

$$u_{HS}(r) = \begin{cases} \infty & (r < \sigma), \\ 0 & (r > \sigma). \end{cases}$$

- Thus, our flat solutions can be conclusively linked to nontrivial interactions (i.e. nonzero $u(r)$).

IV. Conclusions and recommendations

Conclusions and recommendations

- We have studied the existence of flat Ruppeiner metrics, and provided a characterization of both exact and approximate solutions to $R = 0$.
- We found exact, non-ideal gas solutions to $R = 0$ by two approaches: first by specifying metric components, and then by specifying thermodynamic response functions. The second approach addresses a deficiency of the first approach by discarding an unphysical restriction on C_V .
- To more conclusively show that $R = 0$ does not necessarily indicate the absence of interactions, we used the virial expansion of the Helmholtz free energy F , which is closely connected to the intermolecular potential of the system.

Conclusions and recommendations

- This led to the following, surprising result:

There exists no nontrivial B_2 such that both R_V and R_N vanish to lowest order in ρ .

This comprises the main result of this thesis, and suggests that the vanishing of *both* R_V and R_N is a unique feature of the ideal gas.

- Thus, while the vanishing of R_V or R_N alone does not characterize the ideal gas, it seems that the ideal gas is privileged in Ruppeiner's theory by having $R_V = R_N = 0$. To our best knowledge, such an interpretation has not yet been put forward in the literature.

Conclusions and recommendations

- For future work, we hope to either set out an example of a non-ideal gas system where both curvature scalars vanish, or demonstrate rigorously that the ideal gas is indeed uniquely characterized by the property $R_V = R_N = 0$.

Thank you for listening!

Thermodynamics – metric space correspondence

Thermodynamic principles

- (1) An equilibrium state is characterized by r independent state properties $X_1, X_2, \dots, X_r$.
- (2) Derivatives of the thermodynamic potential U further specify an equilibrium state.
- (3) The first law, i.e. U has an exact differential.
- (4) The second law, i.e. U achieves an extremal value for equilibrium states.

Properties of an metric space M

- (i) $\dim(M) = r$
- (ii) $\langle R_i | \lambda R_j + \mu R_k \rangle = \lambda \langle R_i | R_j \rangle + \mu \langle R_i | R_k \rangle$ [distributivity]
- (iii) $\langle R_i | R_j \rangle = \langle R_j | R_i \rangle$ [symmetry]
- (iv) $\langle R_i | R_i \rangle \geq 0$, with equality only if $|R_i\rangle = 0$ [positive-definiteness]